Efficient Pairings Final Exponentiation Using Cyclotomic Cubing for Odd Embedding Degrees Curves

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Abstract

In pairings-based cryptographic applications, final exponentiation with a large fixed exponent ensures distinct outputs for the Tate pairing and its derivatives. Despite notable advancements in optimizing elliptic curves with even embedding degrees, improvements for those with odd embedding degrees, particularly those divisible by **3**, remain underexplored. This paper introduces three methods for applying cyclotomic cubing in final exponentiation and enhancing computational efficiency. The first allows for the execution of one cyclotomic cubing based on the final exponentiation structure. The second leverages some existing seeds structure to enable the use of cyclotomic cubing and extends this strategy to generate new seeds. The third allows generating sparse ternary representation seeds to apply cyclotomic cubing as an alternative to squaring. These optimizations improve performance by up to **19.3%** when computing the final exponentiation for the optimal Ate pairing on $\boldsymbol{BLS}\mathbf{15}$ and $\boldsymbol{BLS}\mathbf{27}$, the target elliptic curves of this study.

Keywords: Elliptic curves, pairings, final exponentiation, cyclotomic cubing, arithmetic.

1 Introduction

Pairings over elliptic curves are crucial for various cryptographic applications, e.g., identity-based encryption [7], short signatures [8], and tri-partite Diffie-Hellman [18]. Significant efforts [5, 6, 19] have been dedicated to developing various families of elliptic curves tailored for pairing applications. Additionally, Additionally, researchers have focused on optimizing the Miller loop [17, 27] and the final exponentiation [15, 20, 25, 26], as these steps account for the majority of the computational complexity in pairings. In [2–4], Barbulescu et al. introduced new parameters that resist an attack on the discrete logarithm problem (DLP) , which was proposed by Kim et al. in [22]. They also demonstrated that, at the 128-security level, the Barreto-Lynn-Scott family of elliptic curves with an embedding degree $k = 12$ ($BLS12$) and the Kachisa-Schaefer-Scott family of elliptic curves with $k = 16$ ($KSS16$) can offer a more efficient pairing than the Barreto-Naehrig family (BN). Barbulescu et al. revealed that elliptic curve families with $k = 9, 15, 27$ might rival $BLS12$, $KSS16$, and (BN).

The final exponentiation consists of computing

$$\frac{p^k - 1}{r} = \frac{p^k - 1}{\Phi_k(p)} \times \frac{\Phi_k(p)}{r}, \text{ where}$$

ϕ_k is the k-th cyclotomic polynomial, the easy part consists of $\frac{p^k-1}{r}$, which is simple to compute, and the hard part is $\frac{\Phi_k(p)}{r}$, which demands effort. The hard part is carried out within a cyclotomic subgroup. The central operation here is an exponentiation by a fixed integer known as the seed. This uses the square-and-multiply (**SM**) method. For elliptic curves with even embedding degrees, such as BLS12, KSS16, and BLS24, cyclotomic squaring is used instead of regular squaring to improve efficiency. This operation is crucial for speeding up the hard part of pairings over these curves. However, cyclotomic squaring is not available for curves with odd embedding degrees, such as $BLS27$. For these curves, **SM** uses standard squaring and multiplication. Granger and Scott [15] found that techniques for cyclotomic squaring could be adapted for curves with odd embedding degrees divisible by 3, leading to *cyclotomic cubing*. Nanjo et al. [24] showed that cyclotomic cubing is 30% faster than regular cubing in $\mathbb{F}_{p^{15}}$ and more efficient than squaring plus multiplication in $\mathbb{F}_{p^{15}}$ and $\mathbb{F}_{p^{27}}$. The structure of the hard part in BLS curves, along with certain seed forms, enables partial use of cyclotomic cubing. This enhances optimization while maintaining the efficiency of the binary representation. According to [24], the cost of the cyclotomic cubing in $\mathbb{F}_{p^{15}}$ is insufficient to apply **CM** to a seed's ternary representation instead of **SM** to its binary representation. Instead, **CM** may be beneficial for seeds with sparse ternary representations. This led us to explore it as an alternative to **SM**, aiming to exploit the efficiency of cyclotomic cubing in pairings over $BLS15$ and $BLS15$.

Our contributions

This paper explores the use of cyclotomic cubing in computing the final exponentiation of the optimal Ate. We propose the following methods:

1. **Direct application of cyclotomic cubing:** This method directly applies cyclotomic cubing by exploiting the hard part's structure in the final exponentiation over BLS curves.
2. **Two consecutive active bits (TCAB):** This method involves searching for a specific pattern of two consecutive active bits in the seed in order to perform cyclotomic cubing. An active bit refers to a **1** in the binary representation of the seed.
3. **Exponentiation using sparse ternary representation:** This method consists in generating new sparse ternary seeds to apply cyclotomic cubing via the **CM** (cubing-and-multiply) method.

Note that these proposals are applicable to any elliptic curve with an odd embedding degree divisible by 3. However, in this paper we focus on the $BLS15$ and $BLS27$ elliptic curves, inspired by improvements in [10]. We place particular emphasis on $BLS27$ due to its suitability for computing the Miller loop and pairing products.

Organization of the paper

This paper is organized as follows: Section 2 establishes the mathematical background, focusing on arithmetic operations in finite fields with extension degrees that are odd divisible by 3. We also recall pairings over $BLS15$ and $BLS27$ elliptic curves. In Section 3, we present our techniques for applying cyclotomic cubing to compute the final exponentiation of the optimal Ate pairing over $BLS15$ and $BLS27$. In Section 4, we evaluate the pairing costs using new seeds presented in ternary representation. Finally, we summarize our findings and suggest future research directions.

Notations

Let $i \in \mathbb{N}^*$. For the remainder of this paper, we adopt the following notations:

- p a big prime number.
- E represents an elliptic curve defined over $\mathbb{F}_p$.
- $\mathbf{M_i}$ indicates the cost of multiplication in $\mathbb{F}_{p^i}$.
- $\mathbf{S_i}$ represents the cost of squaring in $\mathbb{F}_{p^i}$.
- $\mathbf{F_i}$ stands for a Frobenius operation in $\mathbb{F}_{p^i}$
- $\mathbf{I_i}$ signifies the cost of inversion in $\mathbb{F}_{p^i}$.
- $\mathbf{C_{c_i}}$ represents the cost of cyclotomic cubing in $\mathbb{F}_{p^i}$.
- $\mathbf{I_{c_k}}$ signifies the cost of cyclotomic inversion in $\mathbb{F}_{p^i}$.

- For $u \in \mathbb{Z}$, $\mathbf{E_u}$ denotes the cost of exponentiation by u.
- **Sec-level** stands for the security level.

In this paper, we assume that $\mathbf{M_1} \approx \mathbf{S_1}$.

2 Background

Let k be a positive integer such that $3 \mid k$. This section presents the fundamental aspects of arithmetic over $\mathbb{F}_{p^k}$, focusing on cases where k is odd. Additionally, it recalls pairings over elliptic curves with an embedding degree k. For more detailed information, the reader is referred to [1, 4, 10, 21, 24].

2.1 Costs of arithmetic operations over $\mathbb{F}_{p^k}$

We assume that $3 \mid k$, therefore $\mathbb{F}_{p^k}$ is represented as follows:

$$\mathbb{F}_{p^k} = \mathbb{F}_{p^{\frac{k}{3}}}[x]/(x^3 - \xi),$$

where ξ is a cubic non-residue in $\mathbb{F}_{p^{\frac{k}{3}}}$.
The costs of $\mathbf{M_k}$, $\mathbf{S_k}$, $\mathbf{F_k}$, and $\mathbf{I_k}$ are well studied in the literature [1]. Therefore, we will only recall them in Table 1. However, in this section, we will focus on the details of the costs of cyclotomic inversion [10] and cyclotomic cubing [24]. Let us just recall the cyclotomic subgroup.
Definition 1. *The cyclotomic subgroup of $\mathbb{F}_{p^k}$ is given in [12] by:*

$$G_{\Phi_k(p)} = \{\alpha \in \mathbb{F}_{p^k}^*; \alpha^{\Phi_k(p)} = 1\}, \tag{1}$$

where ϕ_k is the $k-th$ cyclotomic polynomial. The order of $G_{\Phi_k(p)}$ is $\Phi_k(p)$.

Since this paper focuses on the fields $\mathbb{F}_{p^{15}}$ and $\mathbb{F}_{p^{27}}$, we recall that Fouotsa et al. demonstrated in [10] that the costs of cyclotomic inversion over $\mathbb{F}_{p^{15}}$ and $\mathbb{F}_{p^{27}}$ are given as follows:

$$\mathbf{I_{c_{15}}} = 3\mathbf{M_5} + 3\mathbf{S_5} \text{ and } \mathbf{I_{c_{27}}} = 3\mathbf{M_9} + 3\mathbf{S_9}.$$

The detailed method for performing this operation is provided in [24]. According to Nanjo et al., the cost of **cyclotomic cubing** over $\mathbb{F}_{p^k}$ is given by:

$$\mathbf{C_{c_k}} = 5\mathbf{M_{\frac{k}{3}}} + 4\mathbf{S_{\frac{k}{3}}} + 3\mathbf{m_{\frac{k}{3},\xi}} + 9\mathbf{A_{\frac{k}{3}}} + \mathbf{a_{\frac{k}{3},1}} + 4\mathbf{h_{\frac{k}{3}}},$$

where $\mathbf{m_{\frac{k}{3},\xi}}$, $\mathbf{a_{\frac{k}{3},1}}$, $\mathbf{h_{\frac{k}{3}}}$, $\mathbf{A'_{\frac{k}{3}}}$, and $\mathbf{A_{\frac{k}{3}}}$ represent the costs of a multiplication by ξ, an addition by 1, a shift operation, a multiplication by 2, and an addition in $\mathbb{F}_{p^{\frac{k}{3}}}$, respectively.
The additions and shift operations are often neglected. We can choose to either include or neglect the multiplication by ξ. If included, the costs of cyclotomic cubing in $\mathbb{F}_{p^{15}}$ and $\mathbb{F}_{p^{27}}$ are as follows:

$$\mathbf{C_{c_{15}}} = 5\mathbf{M_5} + 4\mathbf{S_5} + 3\mathbf{m_{5,\xi}} \quad \text{and} \quad \mathbf{C_{c_{27}}} = 5\mathbf{M_9} + 4\mathbf{S_9} + 3\mathbf{m_{9,\xi}}.$$

Throughout this paper, we will disregard multiplications by ξ throughout this paper for three main reasons. First, several works, including that of Aranha et al. [1], omit multiplications by ξ when evaluating multiplication and squaring costs in $\mathbb{F}_{p^k}$. They use costs derived from previous real implementations. Second, since cyclotomic cubing in $\mathbb{F}_{p^k}$ is comparable to squaring and multiplication, the same assumptions should apply to all three, especially for multiplication by ξ. Finally, in practical scenarios, one can often choose $\xi \in \mathbb{F}_{p^{k/3}}$ such that multiplication by ξ is inexpensive. Therefore, the costs of cyclotomic cubing in $\mathbb{F}_{p^{15}}$ and $\mathbb{F}_{p^{27}}$ are given as follows:

$$\mathbf{C_{c_{15}}} = 5\mathbf{M_5} + 4\mathbf{S_5} \quad \text{and} \quad \mathbf{C_{c_{27}}} = 5\mathbf{M_9} + 4\mathbf{S_9}.$$

In the following Table 1 we provide a summary of operations in $\mathbb{F}_{p^3}$, $\mathbb{F}_{p^9}$, $\mathbb{F}_{p^{15}}$ and $\mathbb{F}_{p^{27}}$.

Fields	Operations	Costs
$\mathbb{F}_{p^3}$	Multiplication $\mathbf{M_3}$	$6\mathbf{M_1}$
	Squaring $\mathbf{S_3}$	$5\mathbf{S_1}$
	Inversion $\mathbf{I_3}$	$37\mathbf{M_1}$
	Fronenius $\mathbf{F_3}$	$2\mathbf{M_1}$
$\mathbb{F}_{p^5}$	Multiplication $\mathbf{M_5}$	$13\mathbf{M_1}$
	Squaring $\mathbf{S_5}$	$13\mathbf{S_1}$
	Inversion $\mathbf{I_5}$	$73\mathbf{M_1}$
	Fronenius $\mathbf{F_5}$	$4\mathbf{M_1}$
$\mathbb{F}_{p^9}$	Multiplication $\mathbf{M_9}$	$36\mathbf{M_1}$
	Squaring $\mathbf{S_9}$	$27\mathbf{M_1}$
	Inversion $\mathbf{I_9}$	$106\mathbf{M_1}$
	Fronenius $\mathbf{F_9}$	$8\mathbf{M_1}$
	Cyclotomic inversion $\mathbf{I_{c_9}}$	$3 \times 6\mathbf{M_1} + 3 \times (2\mathbf{M_1} + 3\mathbf{S_1}) \approx 33\mathbf{M_1}$
	Cyclotomic cubing $\mathbf{C_{c_9}}$	$5 \times 6\mathbf{M_1} + 4 \times 5\mathbf{S_1} \approx 50\mathbf{M_1}$
$\mathbb{F}_{p^{15}}$	Multiplication $\mathbf{M_{15}}$	$78\mathbf{M_1}$
	Squaring $\mathbf{S_{15}}$	$65\mathbf{M_1}$
	Inversion $\mathbf{I_{15}}$	$229\mathbf{M_1}$
	Fronenius $\mathbf{F_{15}}$	$14\mathbf{M_1}$
	Cyclotomic inversion $\mathbf{I_{c_{15}}}$	$3 \times 13\mathbf{M_1} + 3 \times 13\mathbf{S_1} \approx 78\mathbf{M_1}$
	Cyclotomic cubing $\mathbf{C_{c_{15}}}$	$5 \times 13\mathbf{M_1} + 4 \times 13\mathbf{S_1} \approx 117\mathbf{M_1}$
$\mathbb{F}_{p^{27}}$	Multiplication $\mathbf{M_{27}}$	$216\mathbf{M_1}$
	Squaring $\mathbf{S_{27}}$	$153\mathbf{M_1}$
	Inversion $\mathbf{I_{27}}$	$536\mathbf{M_1}$
	Fronenius $\mathbf{F_{27}}$	$26\mathbf{M_1}$
	Cyclotomic inversion $\mathbf{I_{c_{27}}}$	$3 \times 36\mathbf{M_1} + 3 \times (18\mathbf{M_1} + 9\mathbf{S_1}) \approx 189\mathbf{M_1}$
	Cyclotomic cubing $\mathbf{C_{c_{27}}}$	$5 \times 36\mathbf{M_1} + 4 \times (18\mathbf{M_1} + 9\mathbf{S_1}) \approx 288\mathbf{M_1}$

Table 1: The costs of all operations in extension fields $\mathbb{F}_{p^i}$, for $i \in \{3, 5, 9, 15, 27\}$.

2.2 Pairings

Let E be an elliptic curve defined over $\mathbb{F}_p$, r be a large prime factor of $\#E(\mathbb{F}_p)$ and k be the smallest positive integer such that $r \mid (p^k - 1)$. Let $P \in E(\mathbb{F}_p)[r]$ be of order r, and let $f_{r,P}$ be the rational function with the following divisor:

$$Div(f_{r,P}) = r(P) - r(P_\infty).$$

Let $Q \in E(\mathbb{F}_{p^k})[r]$ of order r, and let μ_r denote the group of r-th roots of unity in $\mathbb{F}_{p^k}^*$. The optimal Ate pairing [27] is defined by:

$$e_o : \mathbb{G}_2 \times \mathbb{G}_1 \longrightarrow \mathbb{G}_3$$
$$(Q, P) \longmapsto f_{t-1,Q}(P)^{\frac{p^k-1}{r}}$$

and it can be computed in $\frac{\log_2(r)}{\varphi(k)} + \epsilon(k)$ basic Miller iterations, where

- $\epsilon(k) \leq \log_2(k)$.
- π_p is the Frobenius map that is defined by:
- $\mathbb{G}_1 = E(\overline{\mathbb{F}_p})[r] \cap Ker(\pi_p - 1) = E(\mathbb{F}_p)[r]$.
- $\mathbb{G}_2 = E(\overline{\mathbb{F}_p})[r] \cap Ker(\pi_p - p)$.
- $\mathbb{G}_3 = \{\mu \in \mathbb{F}_{p^k} \mid \mu^r = 1\}$.
- t is the trace of the Frobenius map $\pi_p : E(\overline{\mathbb{F}_p}) \to E(\overline{\mathbb{F}_p})$.

Pairing computation consists of two main stages. The first stage computes the function $f_{t-1,Q}(P)$ using Miller's algorithm [23]. The second stage, known as the final exponentiation, raises this result to the power $\frac{p^k-1}{r}$. This stage is typically divided into two parts: the *easy* part, which can be computed efficiently, and the *hard* part, which is more computationally intensive. Various techniques have been proposed to perform the final exponentiation efficiently [10, 11, 13, 14, 25]. In particular, Zhang et al. [28] addressed the hard part for $k = 27$ by expanding $\frac{\phi_k(p)}{r}$ in base p using a recursive method. Hayashida et al. later generalized this approach to arbitrary embedding degrees by employing a homogeneous cyclotomic polynomial derived from a standard cyclotomic polynomial. In the next section, we examine the computation of the optimal Ate pairing over elliptic curves with embedding degrees 15 and 27.

2.3 Optimal Ate pairing over $BLS15$ and $BLS27$

A BLS curve [6] is a pairing-friendly elliptic curve over a finite field $\mathbb{F}_p$ defined by the equation $y^2 = x^3 + b$, where $b \in \mathbb{F}_p$ is a nonzero constant.

BLS15

The $BLS15$ family consists of parametrized elliptic curves with an embedding degree of 15, defined in [9] by the following parameters:

$$\begin{cases} p = \frac{u^{12}-2u^{11}+u^{10}+u^7-2u^6+u^5+u^2+u+1}{3}, \\ r = u^8 - u^7 + u^5 - u^4 + u^3 - u + 1, \\ t = u + 1, \end{cases}$$

where the seed u is chosen so that p and r are both prime integers, ensuring a secure and efficient $BLS15$ elliptic curve for pairing. The optimal Ate pairing over $BLS15$ is given by:

$$\begin{aligned} e_o \colon \mathbb{G}_2 \times \mathbb{G}_1 &\longrightarrow \mathbb{G}_3 \\ (Q, P) &\longmapsto f_{u,Q}(P)^{\frac{p^{15}-1}{r}}. \end{aligned}$$

The easy part of the final exponentiation is the elevation of Miller's result to the power $(p^5-1)(p^2+p+1)$. The hard part is the result of the easy part power $\frac{\Phi_{15}(p)}{r}$. In [16], for efficiency reasons, they proposed to use a multiple of the hard part of final exponentiation instead of considering the final exponentiation. Note that, an exponent of pairing is a pairing. In this context, they considered $3.\frac{\Phi_{15}(p)}{r}$, where it is developed as follows:

$$3.\frac{\Phi_{15}(p)}{r} = (u-1)^2(u^2+u+1) + \sum_{i=0}^{7} \lambda_i p^i + 3,$$

where $\lambda_7 = 1$, $\lambda_6 = u\lambda_7 - 1$, $\lambda_5 = u\lambda_6$, $\lambda_4 = u\lambda_5 + 1$, $\lambda_3 = u\lambda_4 - 1$, $\lambda_2 = u\lambda_3 + 1$, $\lambda_1 = u\lambda_2$, and $\lambda_0 = u\lambda_1 - 1$.

BLS27

The $BLS27$ family consists of parametrized elliptic curves with an embedding degree of 27, as described in [28] by the following parameters:

$$\begin{cases} r = \frac{u^{18}+u^9+1}{3}, \\ p = (u-1)^2 r + u, \\ t = u + 1. \end{cases}$$

The seed u is selected to guarantee that p and r are prime integers, providing a secure and efficient $BLS27$ elliptic curve for pairing. The optimal Ate pairing over $BLS27$ is given by:

$$\begin{aligned} e_o \colon \mathbb{G}_2 \times \mathbb{G}_1 &\longrightarrow \mathbb{G}_3 \\ (Q, P) &\longmapsto f_{u,Q}(P)^{\frac{p^{27}-1}{r}}, \end{aligned}$$

Raising Miller's result to the power (p^9-1) is known as the easy part of the final exponentiation. However, the elevation of the easy part by $\frac{\Phi_{27}(p)}{r}$ defines the hard part of the final exponentiation and it is developed as follows:

$$\frac{\Phi_{27}(p)}{r} = (u-1)^2(u^2+pu+p^2)(u^6+p^3u^3+p^6)(u^9+p^9+1) + 3.$$

3 Applying cyclotomic cubing in final exponentiation computation

This section demonstrates the applicability of cyclotomic cubing for final exponentiation in two cases: a seed-independent case and a seed-dependent binary representation case.
The final exponentiation costs for BLS15 and BLS27 under **SM** are given in [16] as follows:

$$\mathbf{I_{15}} + 19 \times \mathbf{M_{15}} + \mathbf{S_{15}} + 10 \times \mathbf{F_{15}} + \mathbf{I_{c_{15}}} + 2 \times \mathbf{E_{u-1}} + 9 \times \mathbf{E_u}, \tag{2}$$

and

$$\mathbf{I_{27}} + 9 \times \mathbf{M_{27}} + \mathbf{S_{27}} + 6 \times \mathbf{F_{27}} + 2 \times \mathbf{E_{u-1}} + 17 \times \mathbf{E_u}. \tag{3}$$

3.1 Direct application

For all BLS curves, $\frac{\phi_k(p)}{r}$ is given in [16] in the form $\psi_{k,p,u} + 3$, where $\psi_{k,p,u} \in \mathbb{Z}$. Thus, we can apply cyclotomic cubing to compute the hard part. Let $\alpha \in G_{\Phi_k(p)}$, the computation of the hard part simplifies to $\alpha^{\psi_{k,p,u}+3} = \alpha^{\psi_{k,p,u}}\alpha^{\mathbf{3}}$. The computation of $\alpha^{\mathbf{3}}$ consists in cyclotomic cubing instead of a multiplication and squaring. This results in a fixed computational gain expressed as $(\mathbf{M_k} + \mathbf{S_k}) - \mathbf{C_{c_k}}$. For example:

- for $BLS15$, the gain is:
$$(\mathbf{M_{15}} + \mathbf{S_{15}}) - \mathbf{C_{c_{15}}} \approx 26\mathbf{M_1},$$
- for $BLS27$, the gain is:
$$(\mathbf{M_{27}} + \mathbf{S_{27}}) - \mathbf{C_{c_{27}}} \approx 81\mathbf{M_1}.$$

We will use the above gains to compute the costs of the final exponentiation for $BLS15$ and $BLS27$ using our upcoming methods. These computations will proceed as follows:

$$\mathbf{I_{15}} + 18 \times \mathbf{M_{15}} + \mathbf{C_{c_{15}}} + 10 \times \mathbf{F_{15}} + \mathbf{I_{c_{15}}} + 2 \times \mathbf{E_{u-1}} + 9 \times \mathbf{E_u}, \tag{4}$$

and

$$\mathbf{I_{27}} + 8 \times \mathbf{M_{27}} + \mathbf{C_{c_{27}}} + 6 \times \mathbf{F_{27}} + 2 \times \mathbf{E_{u-1}} + 17 \times \mathbf{E_u} \tag{5}$$

The gain is not significant enough. In the following subsection, we will present a more subtle method for using cyclotomic cubing. This method is called ”Two consecutive active bits”.

3.2 Two consecutive active bits (TCAB)

For some existing seeds in the literature, the binary representation contains two consecutive active bits with a particular form. This allows us to apply cyclotomic cubing in exponentiation within the cyclotomic subgroup. This form can occur in the least, middle, or most significant bits of the seed. Since the first two cases yield no gain, we focus exclusively on the last one.
This subsection explores possible seed forms suitable for **TCAB**, along with existing examples. We also generate new seeds that are useful for the **TCAB** method.

3.2.1 Description

Let h be the Hamming weight of the seed u. Then, u is expressed as:

$$u = 2^{s_1} + 2^{s_2} + \cdots + 2^{s_{h-1}} + 2^{s_h},$$

where $s_1, \ldots, s_h \in \mathbb{N}$ satisfy $s_1 < s_2 < \cdots < s_{h-1} < s_h$. Based on the value of $s_h - s_{h-1}$, we identify two possible cases where we can apply cyclotomic cubing. The first case arises when $s_h - s_{h-1} = 1$, while the second occurs when $s_h - s_{h-1} = 3$. Consequently, the seed u can be expressed as follows:

$$u = 2^{s_1} + 2^{s_2} + \cdots + 2^{s_{h-2}} + 2^{s_{h-1}} \times \mathbf{3^c},$$

where

$$c = \begin{cases} 1, \text{ if } s_h - s_{h-1} = 1, \\ 2, \text{ if } s_h - s_{h-1} = 3. \end{cases}$$

This leads to the following expression for $\alpha \in G_{\phi_k(p)}$:

$$\alpha^u = \alpha^{2^{s_1}+2^{s_2}+\cdots+2^{s_{h-2}}}(\alpha^{\mathbf{2^{s_{h-1}}}})^{\mathbf{3^c}}.$$

Using the current method, computing α^u costs:

$$s_{h-1}\mathbf{S_k} + (h-2)\mathbf{M_k} + c\mathbf{C_{c_k}},$$

whereas using (**SM**) method incurs:

$$s_h\mathbf{S_k} + (h-1)\mathbf{M_k}.$$

3.2.2 Examples using existing seeds

We examined literature seeds where **TCAB** applies to $BLS15$ and $BLS27$ and identified the following:

- For $BLS15$, the found seed is

$$u_{e_{15.190}} = 2^6 + 2^{59} + 2^{62} + \mathbf{2^{73}} + \mathbf{2^{74}}.$$

  $u_{e_{15.190}}$ is proposed in [1] and corresponds to the 190-bit security level.
- For $BLS27$, the identified seeds at the security levels 192 and 256 bits are

$$u_{e_{27.192}} = -2^5 + 2^8 + 2^{12} + 2^{16} + \mathbf{2^{21}} + \mathbf{2^{22}} \quad [4] \quad \text{and} \quad u_{e_{27.256}} = -2^3 + 2^8 + 2^{25} + \mathbf{2^{27}} + \mathbf{2^{28}} \quad [28].$$

The costs of **TCAB** and **SM** applied to the seeds $u_{e_{15.190}}$, $u_{e_{27.192}}$, and $u_{e_{27.256}}$, are given in Table 2.

Seed	Methods	
	TCAB	**SM**
$u_{e_{15.190}}$	$3\mathbf{M_{15}} + 73\mathbf{S_{15}} + \mathbf{C_{c_{15}}} = 5096\mathbf{M_1}$	$4\mathbf{M_{15}} + 74\mathbf{S_{15}} = 5122\mathbf{M_1}$
$u_{e_{27.192}}$	$4\mathbf{M_{27}} + 21\mathbf{S_{27}} + \mathbf{C_{c_{27}}} + \mathbf{I_{c_{27}}} = 4554\mathbf{M_1}$	$5\mathbf{M_{27}} + 22\mathbf{S_{27}} + \mathbf{I_{c_{27}}} = 4635\mathbf{M_1}$
$u_{e_{27.256}}$	$3\mathbf{M_{27}} + 27\mathbf{S_{27}} + \mathbf{C_{c_{27}}} + \mathbf{I_{c_{27}}} = 5256\mathbf{M_1}$	$4\mathbf{M_{27}} + 28\mathbf{S_{27}} + \mathbf{I_{c_{27}}} = 5337\mathbf{M_1}$

Table 2: A cost comparison of **TCAB** and **SM** applied to the Seeds $u_{e_{15.190}}$, $u_{e_{27.192}}$, and $u_{e_{27.256}}$.

The gain obtained is more interesting than that achieved through a direct application of cyclotomic cubing. In order to enhance our method, we propose in the following a generating of new seeds.

3.2.3 Generating new seeds

We aimed to generate new seeds suitable for **TCAB** while taking into account the following constraints:

1. Following the recommendations of Barbulescu and Duquesne [2] for discrete logarithm computation over the field $\mathbb{F}_{p^k}$ concerning the size of p^k.
2. Generating binary seeds to avoid additional cyclotomic inversion costs.
3. Produce odd seeds or those whose least significant active bit is equal to 2, guaranteeing $\mathbf{E_{u-1}} \leq \mathbf{E_u}$.
4. Generating more efficient seeds comparing to those proposed in [4, 10].

For each security level, the seed is designed to match the size and Hamming weight of its counterpart in [4] or [10]. Additionally, the **TCAB** requires the last two active bits to be fixed which reduces the search space. In some cases, we cannot find a seed suitable for **TCAB** for a given size and Hamming weight. When this occurs, these parameters must be slightly adjusted, while ensuring compliance with the security constraints outlined in [2]. Through the pursuit of a balance between efficiency and security, we identified the optimal seeds that are suitable for **TCAB**. Table 3 presents the proposed seeds, systematically arranged in ascending order of security level. It also includes the degree of embedding of the corresponding curve, the sizes of the primes r and p, and the coefficient b of the equation of the curve. Table 4 reports the final

Seed	Sec-level	k	Size(r)	Size(p)	Size(p^k)	b	DL algorithm
$1 + 2^{16} + 2^{21} + 2^{27} + 2^{28}$	128	15	229	342	5123	1	SexTNFS
$2 + 2^{9} + 2^{12} + 2^{15}$		27	272	303	8160	16	SexTNFS
$2^{17} + 2^{20} + 2^{68} + 2^{70} + 2^{71}$	192	15	574	859	12883	6	SexTNFS
$1 + 2^{11} + 2^{20} + 2^{23} + 2^{24}$		27	443	492	13265	2	SexTNFS
$2 + 2^{41} + 2^{45} + 2^{48}$	256	27	866	963	25975	3	exTNFS

Table 3: New valid seeds for **TCAB** use.

exponentiation costs for $BLS15$ and $BLS27$, obtained by applying the **TCAB** to the generated seeds and using the expressions (4) and (5).

Seed	Sec-level	k	Cost
$1+2^{16}+2^{21}+2^{27}+2^{28}$	128	15	$24978\mathbf{M_1}$
$2+2^{9}+2^{12}+2^{15}$		27	$56744\mathbf{M_1}$
$2^{17}+2^{20}+2^{68}+2^{70}+2^{71}$	192	15	$56191\mathbf{M_1}$
$1+2^{11}+2^{20}+2^{23}+2^{24}$		27	$86921\mathbf{M_1}$
$2+2^{41}+2^{45}+2^{48}$	256	27	$152675\mathbf{M_1}$

Table 4: The final exponentiation costs for $BLS15$ and $BLS27$ using the new seeds.

The following challenges were encountered during the search for new seeds suitable for **TCAB**:

- For $k = 27$ at 256-bit security, we could not find a SexTNFS-secure seed more efficient than that of [10]. Our generated seed is only exTNFS secure.
- Although it is possible to generate a 256-bit seed for the $BLS15$ curve, doing so requires a large seed size, which negatively impacts the efficiency of the Miller loop.

3.2.4 Comparison

The comparison proceeds in two phases: first, assessing the efficiency of **TCAB** versus **SM** on the new seeds; second, comparing **TCAB** on the new seeds with **SM** on the seeds proposed in [4, 10].
We use the expressions (2) and (3) and the Table 1 to compare, in Table 5, the final exponentiation cost over $BLS15$ and $BLS27$ elliptic curves when applying **TCAB** and **SM**. Additionally, we evaluate **TCAB**'s gain over **SM**.

Seed	Sec-level	k	Method	Cost	Gain (TCAB/SM)
$1+2^{16}+2^{21}+2^{27}+2^{28}$	128	15	TCAB	$24978\mathbf{M_1}$	$312\mathbf{M_1}$
			SM	$25290\mathbf{M_1}$	
$2+2^{9}+2^{12}+2^{15}$		27	TCAB	$56744\mathbf{M_1}$	$1962\mathbf{M_1}$
			SM	$58706\mathbf{M_1}$	
$2^{17}+2^{20}+2^{68}+2^{70}+2^{71}$	192	15	TCAB	$56191\mathbf{M_1}$	$312\mathbf{M_1}$
			SM	$56503\mathbf{M_1}$	
$1+2^{11}+2^{20}+2^{23}+2^{24}$		27	TCAB	$86921\mathbf{M_1}$	$1085\mathbf{M_1}$
			SM	$88006\mathbf{M_1}$	
$2+2^{41}+2^{45}+2^{48}$	256	27	TCAB	$152675\mathbf{M_1}$	$1962\mathbf{M_1}$
			SM	$154637\mathbf{M_1}$	

Table 5: Comparison of the cost of the final exponentiation for pairings over $BLS15$ and $BLS27$ applying **TCAB** and **SM** to the new seeds.

Table 5 indicates that the newly generated seeds demonstrate greater efficiency under the **TCAB** setting compared to **SM**.
We compare, in Table 6, final exponentiation costs for optimal Ate pairing on $BLS15$ and $BLS27$, applying **TCAB** to new seeds and **SM** to existing ones [4, 10].

Seed	Sec-level	k	Complexity	Gain
$1+2^{16}+2^{21}+2^{27}+2^{28}$ (**This work**)	128	15	$\mathbf{I_{15}}+49\mathbf{M_{15}}+297\mathbf{S_{15}}+12\mathbf{C_{c_{15}}}+\mathbf{I_{c_{15}}}+10\mathbf{F_{15}}=24978\mathbf{M_1}$	$1911\mathbf{M_1}(7\%)$
$2^2+2^5+2^{19}+2^{31}$ [10]			$\mathbf{I_{15}}+54\mathbf{M_{15}}+342\mathbf{S_{15}}+\mathbf{I_{c_{15}}}+10\mathbf{F_{15}}=26889\mathbf{M_1}$	
$2+2^9+2^{12}+2^{15}$ (**This work**)	128	27	$\mathbf{I_{27}}+46\mathbf{M_{27}}+228\mathbf{S_{27}}+39\mathbf{C_{c_{27}}}+6\mathbf{F_{27}}=56744\mathbf{M_1}$	$2772\mathbf{M_1}(4.6\%)$
$2^3+2^4+2^{11}+2^{15}$ [4]			$\mathbf{I_{27}}+68\mathbf{M_{27}}+286\mathbf{S_{27}}+2\mathbf{I_{c_{27}}}+6\mathbf{F_{27}}=59516\mathbf{M_1}$	
$2^{17}+2^{20}+2^{68}+2^{70}+2^{71}$ (**This work**)	192	15	$\mathbf{I_{15}}+53\mathbf{M_{15}}+770\mathbf{S_{15}}+12\mathbf{C_{c_{15}}}+3\mathbf{I_{c_{15}}}+10\mathbf{F_{15}}=56191\mathbf{M_1}$	$559\mathbf{M_1}(1\%)$
$1+2^5+2^9+2^{40}+2^{72}$ [10]			$\mathbf{I_{15}}+61\mathbf{M_{15}}+793\mathbf{S_{15}}+\mathbf{I_{c_{15}}}+10\mathbf{F_{15}}=56750\mathbf{M_1}$	
$1+2^{11}+2^{20}+2^{23}+2^{24}$ (**This work**)	192	27	$\mathbf{I_{27}}+63\mathbf{M_{27}}+437\mathbf{S_{27}}+20\mathbf{C_{c_{27}}}+6\mathbf{F_{27}}=86921\mathbf{M_1}$	$4527\mathbf{M_1}(5\%)$
$1+2^4+2^{14}+2^{17}+2^{25}$ [10]			$\mathbf{I_{27}}+83\mathbf{M_{27}}+476\mathbf{S_{27}}+6\mathbf{F_{27}}=91448\mathbf{M_1}$	
$2+2^{41}+2^{45}+2^{48}$ (**This work**)	256	27	$\mathbf{I_{27}}+46\mathbf{M_{27}}+855\mathbf{S_{27}}+39\mathbf{C_{c_{27}}}+6\mathbf{F_{27}}=152675\mathbf{M_1}$	$14355\mathbf{M_1}(8.6\%)$
$1+2^9+2^{28}+2^{42}+2^{51}$ [10]			$\mathbf{I_{27}}+83\mathbf{M_{27}}+970\mathbf{S_{27}}+6\mathbf{F_{27}}=167030\mathbf{M_1}$	

Table 6: Comparison of our seeds and existing ones based on final exponentiation cost over $BLS15$ and $BLS27$.

We successfully generated, in this subsection, new seeds suitable for the application of **TCAB**. These seeds enabled the use of cyclotomic cubing, leading to optimized final exponentiation computation over the $BLS15$ and $BLS27$ curves.
In the previous section, we used the seed's binary representation to apply cyclotomic cubing and optimize the final exponentiation. Next, we explore generating seeds with sparse ternary representations, potentially better suited for cyclotomic cubing .

4 Exponentiation using sparse ternary representation

We focus, in this section, on enhancing the computation of the final exponentiation in optimal Ate pairing over $BLS15$ and $BLS27$ by generating sparse ternary seeds and using cyclotomic cubing.

4.1 Cubing and multiplication (CM)

To benefit from using ternary representation, we introduce an alternative to **SM** that replaces squaring with cubing. Given a seed u, its ternary representation is:

$$tern(u) = (t_0 t_1 \cdots t_{n-1})_3,$$

where $t_i \in \{0, 1, 2\}$, and

$$u = \sum_{i=0}^{n-1} t_i 3^i,$$

with n denoting its length. Let $k \in \mathbb{N}^*$ such that $3 \mid k$ and $\alpha \in G_{\Phi_k(p)} \subset \mathbb{F}_{p^k}$. To exploit cyclotomic cubing and ternary sparsity, α^u is computed using 'cubing and multiply' (**CM**), detailed in Algorithm 1. This method applies one cubing per digit and multiplies when the digit is nonzero.

4.2 Generating new sparse ternary seeds

Our objective is to construct new sparse ternary seeds that satisfy the following constraints:

1. Following the security requirements outlined in [2], which address the size of p^k.
2. Each new seed must have its least significant bit set to 1, ensuring the reduction of the cost $\mathbf{E_{u-1}}$.
3. The newly proposed seeds should maintain competitiveness in terms of final exponentiation cost when compared to the seeds presented in [10].

We denote our seeds by u_t and those of [10] by u_b. Let h_t and h_b be the ternary and binary Hamming weights of u_t and u_b. Exponentiation in $G_{\phi_k(p)}$ by u_t using **CM** costs:

$$c_t = (h_t - 1)\mathbf{M_k} + \mathbf{S_k} + (\log_3(u_t) - 1)\mathbf{C_{c_k}}.$$

Algorithm 1 CM (Cubing and multiplication)

Input: Parameter $u = (t_0, t_1, \cdots, t_n)_3$, $\alpha \in \mathbb{F}_{p^k}$
Output: α^u.

1. $r = 1$,
2. $\beta = \alpha^2$,//**If the ternary representation of** u **contains** 2
3. **for** $j = n - 1$ **down to** 0 **do**
   3.1 $r \leftarrow r^3$,
   3.2 **if** $t_j = 1$ **then** $r \leftarrow r\alpha$,
   3.3 **if** $t_j = 2$ **then** $r \leftarrow r\beta$.
4. **return** r.

With **SM**, exponentiation by u_b in $G_{\phi_k(p)}$ costs:

$$c_b = (h_b - 1)\mathbf{M_k} + (\log_2(u_b) - 1)\mathbf{S_k}.$$

By satisfying the condition $c_t < c_b$, we successfully generated the following seeds:

- **128-bit security level:** For the curve BLS15, the seed is

$$1 + 3^2 + 3^5 + 3^{10} + 3^{16}.$$

- **192-bit security level:** For the curve BLS15, the seed is

$$1 + 3^{21} + 3^{28} + 3^{29} + 2 \times 3^{32}.$$

  For the curve BLS27, the seed is

$$1 + 2 \times 3^9 + 3^{11}.$$

- **256-bit security level:** For the curve BLS27, the seed is

$$1 + 2 \times 3^{12} + 2 \times 3^{15} + 2 \times 3^{17} + 2 \times 3^{21}.$$

4.3 Comparison

In this section, we conduct the comparison as follows:

1. Table 7 highlights the security properties of our new ternary seeds versus those proposed in [10].

Seed	Sec-level	k	Size(p)	Size(r)	Size($\mathbf{p^k}$)	DL Alg
$1 + 3^2 + 3^5 + 3^{10} + 3^{16}$	128	15	303	203	4542	exTNFS
$2^2 + 2^5 + 2^{19} + 2^{31}$ [10]			371	249	5557	SexTNFS
$1 + 3^{21} + 3^{28} + 3^{29} + 2 \times 3^{32}$	192	15	620	415	9292	exTNFS
$1 + 2^5 + 2^9 + 2^{40} + 2^{72}$ [10]			863	577	12937	SexTNFS
$1 + 2 \times 3^9 + 3^{11}$	192	27	353	318	9529	exTNFS
$1 + 2^4 + 2^{14} + 2^{17} + 2^{25}$ [10]			511	410	13461	SexTNFS
$1 + 2 \times 3^{12} + 2 \times 3^{15} + 2 \times 3^{17} + 2 \times 3^{21}$	256	27	685	616	18482	exTNFS
$1 + 2^9 + 2^{28} + 2^{42} + 2^{51}$ [10]			1019	883	27499	SexTNFS

Table 7: Comparison of security properties of our ternary seeds with binary seeds in [10].

2. Table 8 compares the cost of exponentiation in $G_{\phi_{15}(p)}$ and $G_{\phi_{27}(p)}$ applying **CM** to our new ternary seeds and **SM** to those proposed in [10].

Seed	k	Sec-level	Method	Cost	Gain
$1+3^2+3^5+3^{10}+3^{16}$	15	128	CM	$4\mathbf{M_{15}}+16\mathbf{C_{c_{15}}}=2184\mathbf{M_1}$	$65\mathbf{M_1}$
$2^2+2^5+2^{19}+2^{31}$ [10]			SM	$3\mathbf{M_{15}}+31\mathbf{S_{15}}=2249\mathbf{M_1}$	
$1+3^{21}+3^{28}+3^{29}+2\times 3^{32}$	15	192	CM	$4\mathbf{M_{15}}+\mathbf{S_{15}}+32\mathbf{C_{c_{15}}}=4121\mathbf{M_1}$	$871\mathbf{M_1}$
$1+2^5+2^9+2^{40}+2^{72}$ [10]			SM	$4\mathbf{M_{15}}+72\mathbf{S_{15}}=4992\mathbf{M_1}$	
$1+2\times 3^9+3^{11}$	27	192	CM	$2\mathbf{M_{27}}+\mathbf{S_{27}}+11\mathbf{C_{c_{27}}}=3753\mathbf{M_1}$	$936\mathbf{M_1}$
$1+2^4+2^{14}+2^{17}+2^{25}$ [10]			SM	$4\mathbf{M_{27}}+25\mathbf{S_{27}}=4689\mathbf{M_1}$	
$1+2\times 3^{12}+2\times 3^{15}+2\times 3^{17}+2\times 3^{21}$	27	256	CM	$4\mathbf{M_{27}}+\mathbf{S_{27}}+21\mathbf{C_{c_{27}}}=7065\mathbf{M_1}$	$1602\mathbf{M_1}$
$1+2^9+2^{28}+2^{42}+2^{51}$ [10]			SM	$4\mathbf{M_{27}}+51\mathbf{S_{27}}=8667\mathbf{M_1}$	

Table 8: Comparison of the cost of exponentiation by the ternary seeds and the seeds of [10] in $G_{\phi_{15}(p)}$ and $G_{\phi_{27}(p)}$.

3. Given the positive gains shown in Table 8, we extend the comparison to the final exponentiation cost of the optimal Ate pairing on the $BLS15$ and $BLS27$ curves. Specifically, we apply **CM** to our newly generated seeds and **SM** to the seeds from [10], and, using formulas ((4))–((3)), we present the resulting final exponentiation costs in Table 9.

Seed	k	Sec-level	Cost	Gain
$1+3^2+3^5+3^{10}+3^{16}$ (**This work**)	15	128	$25836\mathbf{M_1}$	$1053\mathbf{M_1}(3.9\%)$
$2^2+2^5+2^{19}+2^{31}$ [10]			$26889\mathbf{M_1}$	
$1+3^{21}+3^{28}+3^{29}+2\times 3^{32}$ (**This work**)	15	192	$47143\mathbf{M_1}$	$9607\mathbf{M_1}(16.9\%)$
$1+2^5+2^9+2^{40}+2^{72}$ [10]			$56750\mathbf{M_1}$	
$1+2\times 3^9+3^{11}$ (**This work**)	27	192	$73583\mathbf{M_1}$	$17865\mathbf{M_1}(19.3\%)$
$1+2^4+2^{14}+2^{17}+2^{25}$ [10]			$91448\mathbf{M_1}$	
$1+2\times 3^{12}+2\times 3^{15}+2\times 3^{17}+2\times 3^{21}$ (**This work**)	27	256	$136511\mathbf{M_1}$	$30519\mathbf{M_1}(18.2\%)$
$1+2^9+2^{28}+2^{42}+2^{51}$ [10]			$167030\mathbf{M_1}$	

Table 9: Comparison of the final exponentiation costs and the gain offered by our seeds

Though slightly less secure than those in [10], our seeds are more efficient and remain exTNFS-secure.

5 Conclusion

This paper presents the first application of cyclotomic cubing to the computation of the final exponentiation cost in the Ate pairing on the $BLS15$ and $BLS27$ curves. We began with a straightforward application that exploits the structure of the final exponentiation expression for BLS curves, resulting in an explicit formula for estimating the final exponentiation cost for these curves. Building on this, we introduced the **TCAB** method, which leverages cyclotomic cubing through a specific structure in the seed's binary representation. To further advance this approach, we generated novel seeds with sparse ternary representations aimed at improving efficiency in the final exponentiation step for $BLS15$ and $BLS27$. While these new seeds achieve slightly lower security levels compared to existing constructions, they offer greater computational efficiency. A notable challenge, however, is that a sparse ternary representation does not necessarily imply sparsity in the binary form, which can reduce the effeciency of Miller's algorithm due to its reliance on binary sparsity. As an open problem, we propose investigating seeds that maximize the number of cyclotomic cubings while preserving the efficiency of the Miller algorithm. Additionally, it would be valuable to design a variant of the Miller algorithm based on a "tripling-and-add" strategy.

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